

Grailcard

Technical Implementation & Competitive Analysis

Version 0.1 (Milestone 1 + Identification) · August 18, 2026 · Internal report

1. Executive Summary

Grailcard is a trading-card pre-grading system built on a single thesis: in a market where every competitor advertises unfalsifiable accuracy percentages, the defensible position is **demonstrated trustworthiness** — refuse bad input instead of guessing from it, output measurements instead of opinions, attach a confidence figure to every claim, and say plainly when something cannot be measured.

This document describes what has been implemented, how each stage works, the measured accuracy of the system against ground truth, and a same-image, head-to-head comparison against CardGrader.AI — the most instructive result in this report. On identical input, Grailcard produced a small number of verifiable claims (correct identification, a centering measurement with a disclosed low-confidence flag, and a market price consistent with live exchange data), while CardGrader.AI produced a detailed, authoritative-looking report containing a misidentification worth a ~300× valuation error, sub-grades for physical defects on an image of a card that does not physically exist, and internally contradictory price estimates.

2. Design Thesis and Market Context

The AI pre-grading market (PreGradeCards, CardGrade.io, CardMintAI, TCGrader, ZeroPop, CardGrading.app, SnapGradeAI, CardGrader.AI, and others) is crowded and priced to the floor: entry plans near \$4.99/month, per-card pricing around \$0.25, free scans used for acquisition. Vendors self-report 87–95% accuracy on their own test sets; independent observation puts photo-based AI grading at roughly 70–85% agreement with PSA, usually "within one grade" — a tolerance that is commercially meaningless at the top of the scale, where the difference between a 9 and a 10 is the entire economic question.

Grailcard's differentiation is structural, not incremental:

- **Refuse bad input.** A quality gate rejects blurry, glared, small, or low-resolution photos with a specific retry instruction, at no charge — instead of returning a number from data that cannot support one.
- **Measure, don't opine.** Centering is computed geometrically from the printed border and reported as ratios (e.g. 52/48), with the measured border lines drawn on the image so the user can see exactly where the measurement was taken.
- **Report uncertainty.** Every measurement carries a confidence score derived from signal quality, and it is shown to the user — including when it is low.
- **Honest non-answers.** Borderless/full-art cards return "centering not measurable" rather than a fabricated ratio. Claims the system cannot yet make (corners, edges, surface) are absent, not invented.

3. System Architecture

Component	Stack	Responsibility
apps/web (port 3000)	Next.js 15, React 19	Scanner UI: front/back capture slots with corner brackets, animated scan sweep, result report (identity, detection-line viewer, centering, valuation), recent scans.
apps/api (port 8180)	NestJS, Node 24 built-in SQLite, local disk storage	Scan orchestration and persistence; calls the vision service per capture; catalog identification + valuation via TCGdex; serves stored images.
services/vision (port 8100)	Python 3.12, FastAPI, OpenCV, RapidOCR (ONNX)	All computer vision: detection & perspective normalization, quality gate, centering measurement, on-card text reading. Stateless: image in, JSON out.
packages/shared	TypeScript + Zod	The scan-record contract shared by web and API and mirrored by the vision service.

All computer vision is in-house. No third-party vision API is used in production — a deliberate decision (\$7): building on a vendor such as Ximilar (~€0.06–0.10 per graded scan) caps accuracy at the vendor's model, makes the product a reseller of calls every competitor can buy, and forecloses the differentiators above, none of which a generic API exposes. External services are limited to **commodity data**: the free TCGdex catalog (identification + market prices), fetched server-side with graceful degradation.

4. Pipeline Implementation

4.1 Detection and normalization (`detect.py`)

The card quad is located with a two-strategy contour search: Canny edges (dilated) first, then an Otsu binarization fallback that still separates a light card from a darker background when heavy blur destroys the edge response — so a blurry photo is *found* and then rejected for the true reason (blur) rather than misreported as "no card". Candidate contours are filtered by area (2–95% of frame) and aspect ratio (0.55–0.90 short/long, around the 63×88 mm card's 0.716), then perspective-warped to a canonical 750×1050 crop. Landscape shots are auto-rotated.

4.2 Quality gate (`quality.py`)

Applied most-fundamental-failure-first; every rejection carries a plain-language reason and a concrete retry hint. All thresholds live in a single config file.

Check	Method	Threshold	Rejection
Card present	Quad search (above)	—	card_not_found
Card size in frame	Quad area / frame area	$\geq 10\%$	card_too_small
Resolution	Card short side in source pixels	≥ 450 px	resolution_too_low
Focus	Laplacian variance on warped crop	≥ 40	too_blurry
Glare	Specular clusters (HSV $V \geq 250$, $S \leq 40$, connected components)	$\leq 4\%$ of card	too_much_glare , hint names the glare's location

4.3 Centering measurement (`centering.py`)

On the canonical crop the card's outer edge *is* the image edge, so centering reduces to locating the inner print border on each side. For each side, the mean absolute Sobel gradient is profiled across the central 70% band (corners excluded) within a search window covering the outer 25% of the card; the dominant peak marks the border. Ratios are pure pixel arithmetic. The peak's strength relative to the profile median yields both a **measurability verdict** (ratio ≥ 3 required — borderless cards honestly fail this) and a **confidence score**. PSA pass verdicts apply published standards: front $\leq 60/40$ for a 10, $\leq 65/35$ for a 9; back $\leq 75/25$ and $\leq 90/10$ when a back capture is provided. An annotated overlay (the four measured border lines drawn on the card) is returned with every measurement.

4.4 Identification (`identify.py` + API `tcgdex.ts`)

RapidOCR (offline ONNX; no per-call cost) reads the warped front. The title is taken from prominent text in the top band, cleaned of OCR artifacts (merged spaces are split on case boundaries; mangled "ex/GX" logo fragments are stripped); the collector number is regex-matched in the bottom band. The API then queries the free TCGdex catalog with progressive seeds and scores candidates by character-bigram Dice similarity, with a bonus when the printed collector number agrees; matches below 0.45 are discarded rather than guessed. Identification intentionally runs even when the grading gate rejects the photo — reading a title needs far less resolution than measuring a border, so a rejected photo still tells the user which card was seen and what it is worth. The UI always shows the raw OCR text and the match percentage.

4.5 Valuation

TCGdex card records carry live TCGplayer (USD low/mid/high/market, per print variant) and Cardmarket (EUR low/trend/30-day average) prices with update timestamps; these are displayed explicitly labeled **raw / ungraded**, with the price date. The production plan replaces per-request fetching with nightly snapshot ingestion into Postgres (PriceCharting for graded-slab prices, JustTCG for raw condition-specific prices), keeping data costs flat ($< \$60/\text{month}$) and independent of scan volume while accumulating proprietary price history.

5. Verification and Measured Accuracy

The vision pipeline is verified by a golden-fixture suite (20 tests, all passing) built on a synthetic generator that renders a card with **exactly known** border ratios, rotated 8° like a handheld shot, then composited and re-measured through the full pipeline. Because ground truth is exact, accuracy claims below are measurements, not estimates.

True L/R	True T/B	Measured L/R	Measured T/B	Error L/R (pp)	Error T/B (pp)
50/50	50/50	49.5/50.5	49.6/50.4	0.5	0.4
60/40	40/60	58.7/41.3	40.0/60.0	1.3	0.0
40/60	60/40	39.1/60.9	58.9/41.1	0.9	1.1
55/45	45/55	53.6/46.4	45.4/54.6	1.4	0.4
35/65	50/50	35.5/64.5	49.6/50.4	0.5	0.4
65/35	55/45	63.6/36.4	54.6/45.4	1.4	0.4
58/42	50/50	56.4/43.6	49.6/50.4	1.6	0.4
64/36	50/50	62.7/37.3	49.6/50.4	1.3	0.4
72/28	50/50	70.0/30.0	49.6/50.4	2.0	0.4

Mean absolute error 0.8 percentage points; maximum 2.0. PSA pass/fail verdicts were correct in all cases, including the deliberately borderline 58/42 (passes 10) and 64/36 (9 only). The gate tests verify each rejection reason fires for its true cause, and that a borderless card returns "not measurable" rather than a number. End-to-end over HTTP (web → API → vision → SQLite → report), a front/back pair measured 59/41 front (true 60/40) and 68/32 back (true 70/30), with the combined PSA verdict correctly applying the looser back standard.

Disclosed limitation — modern thin-border cards.

On a modern ME-era card (Mega Slowbro ex) with a very thin silver frame, the border scan locked onto an artwork edge and reported 23/77 where truth is ~50/50. Classic-border cards measure correctly. Per-era border calibration is the top roadmap item; until then the product's own confidence score is the guardrail — thin, ambiguous border peaks produce low confidence, and that figure is always displayed.

6. Case Study: Same Image, Grailcard vs CardGrader.AI

A single test image — the official Base Set Charizard #4 render (Unlimited edition), composited onto a dark background at a slight angle — was submitted to both systems on August 18, 2026. The image is perfectly centered by construction and depicts **no physical card at all**.

Dimension	Grailcard	CardGrader.AI
Identification	Charizard, Base Set #4, Rare — correct edition; OCR text and 100% match score shown.	"1999 Base Set 1st Edition Charizard Holo 4/102" — the render carries no 1st Edition stamp. This single error drives the entire valuation narrative.
Condition claims	Centering only: 52/48 L/R, 41/59 T/B, confidence 27% displayed . No corner/edge/surface claims — the system does not measure them yet, so it says nothing.	CG 9.3 with sub-grades (Centering 9.1, Corners 9.3, Edges 10, Surface 10) and a specific defect narrative ("one corner on the back bottom-left is not perfectly crisp") — physical-condition findings, including a back analysis, for a digital image with no physical card and no back photo.
Valuation	\$852.43 TCGplayer market / €312.69 Cardmarket trend, labeled raw/ungraded, dated — consistent with live exchange data for the Unlimited card.	Raw baseline \$222.61 from eBay comps the report itself describes as "heavily polluted with counterfeit 'PRXY' and custom-made cards"; graded estimates PSA 9 = \$67,000 while PSA 10 = \$10,000 (a 10 priced at 15% of a 9); headline "Potential ROI 29,997%" beside the recommendation "Don't Grade It"; a CG 9.3 condition verdict valued "conservatively" at a PSA 6–7.
Input scrutiny	Passed the gate (high-resolution, sharp, no glare) — see limitation below.	None apparent.

What the comparison actually shows.

The failure mode is not "their model is less accurate." It is architectural: a system designed to always deliver a rich answer will fabricate whatever the answer requires — an edition it cannot verify, corner wear that does not exist, a coherent-sounding narrative wrapped around incoherent numbers. A misidentified edition alone turned a ~\$850 card into a "\$67,000" one. Grailcard's design makes fewer claims, and each is traceable: the OCR text is shown, the measured border lines are drawn on the image, the confidence is printed, and the capabilities it lacks are simply absent from the report.

Grailcard's own errors in this test, disclosed.

(1) The T/B measurement of 41/59 on a perfectly centered render is ~9pp off — the composite's JPEG compression and the render's border profile produced weak gradient peaks, and the system flagged exactly that by reporting 27% confidence; the number should be read accordingly. (2) More fundamentally, **both** systems analyzed a picture of a digital image as though it were a physical card. Grailcard should detect and refuse screen photos, prints, and renders (no print grain, no paper texture); this authenticity gate is on the roadmap and is precisely the kind of refusal the product's thesis demands.

7. Competitive Landscape

Competitor	Model	Pricing signal (2026)	Structural position
CardGrader.AI	Photo app + eBay comps + conversational "Carl"; grade, ROI, listing automation	Freemium; eBay affiliate revenue	Maximal-answer design: always returns full grade + valuation narrative (see §6).
PreGradeCards, CardGrade.io, TCGrader, ZeroPop, SnapGradeAI, CardGrading.app, et al.	Photo → PSA-style 1–10 estimate	~\$4.99/mo entry; ~\$0.25/card; one vendor offers unlimited \$10/mo	Undifferentiated; compete on self-reported accuracy (87–95% on internal sets).
CardGrade "Forensic" tier	Guided per-corner capture, premium price	Premium per-card	Validates that capture quality is the real bottleneck; also offers a 2-grade credit-back guarantee.
Ximilar	B2B vision API (identify, grade, centering)	Metered credits: full grade ≈ €0.06–0.10/scan at plan rates	Supplies the "generic wrapper" tier; does not store images; no input gate, bands, or confidence surface.
TAG / AGS	Mail-in machine grading (physical scanners)	Per-slab fees, days–weeks turnaround	High accuracy, zero instant-answer utility; complementary rather than competitive.

Grailcard's cost structure against this field: all vision in-house at ~\$0.003 marginal cost per scan (compute + storage), identification and valuation on free catalog data, and a planned flat <\$60/month for premium price feeds — versus €0.06–0.10/scan perpetual COGS and a shared accuracy ceiling for every wrapper built on the same vendor API.

8. Current Limitations

- **Centering only.** Corners, edges, and surface are not yet analyzed; the product makes no claims about them. (Corner-macro capture and trained per-criterion models are Phases 3–4.)
- **Modern thin-border cards** can mislead the border scan (§5); low confidence is the current guardrail, per-era calibration the fix.
- **No authenticity / digital-source detection.** A high-quality render or screen photo passes the gate (§6). Planned: texture/print-grain analysis to refuse non-physical sources.
- **Valuation is raw-market only.** Graded-slab prices, population reports, and the submit-or-not expected-value calculation await the paid data integrations.

- **Identification is OCR-driven.** Extremely low-resolution or non-Latin cards can fail to match; the embedding-based visual index (DINOv2 + pgvector) is the Phase 2 upgrade.

9. Roadmap

1. Per-era border calibration keyed off identification (fixes modern thin-border centering).
2. Authenticity gate: refuse renders, screen photos, and prints.
3. Live guided capture in-browser (corner brackets with real-time quality checks before the shot).
4. Embedding-based identification index; corner-macro captures; corner/edge analyzers.
5. Graded-price data + population reports → grade-band probabilities and submission ROI.
6. Public prediction-vs-actual dashboard verified against PSA cert numbers — the falsifiable accuracy claim no competitor currently publishes.

Grailcard internal technical report, August 18, 2026. Accuracy figures in §5 are reproducible via `npm run test:vision` (20 tests) and the synthetic generator in `services/vision/tests/synth.py`. CardGrader.AI output in §6 quoted from a user session of August 18, 2026 on the same test image; market context (§2, §7) from vendor sites and public documentation retrieved August 2026 — pricing in this space changes monthly and should be re-verified before external use. Market data: TCGdex (tcgdex.dev), prices updated August 17, 2026.